Analysis of the Earth's Supply of Crude Oil

MATH 222 – Introduction to Dynamical Systems

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Defining the Problem

Crude oil is arguably one of the most important natural resources in modern times. It has fueled innovation and economic growth for centuries and now is used in a wide variety of applications. From cosmetic supplies to plastic to nylon to gasoline, this substance is incredibly versatile. Our society has become increasingly reliant on petroleum, but given its finite nature it is crucial that we know how long it will last.

Between issues with pinpointing the exact supply of crude oil and the highly dynamic nature of its consumption, determining its lifespan is quite difficult. Thus, implementing a mathematical model will help account for the relevant factors as they change over time, ultimately leading to a reasonable estimation of its lifespan. We will be forced to make assumptions to create this model, but by taking care in such we can minimize the error that ensues.

Determining how long the Earth's crude oil will last has two major components. First, estimating the total supply available for use, and second, modeling consumption of this resource over time. Although there are fairly accurate estimations of how much crude oil exists, these numbers are still subject to quite a bit of error. Furthermore, the development of international politics and other socioeconomic issues over time may get in the way of being able to fully utilize this resource, and the impact of these factors is especially hard to predict. Regarding consumption, dynamic modeling techniques will be used to account for changes in trends over time. This should address issues including, but not limited to, increased popularity of electric vehicles, improvements in fuel efficiency, popularity of public transportation, changes in transportation needs, and more. Accounting for the plethora of factors that affect crude oil's lifespan and making the right assumptions, so that error is minimal, are the key aspects in creating an accurate model of this resource.

Part One: Preliminary Analysis

Given the complexity of this problem, an initial rough estimate will serve as a starting point before creating a detailed model. In doing so, we will assume that:

- The volume of the Earth's crust represents the total supply of crude oil
    - The volume, assuming Earth is perfectly spherical, can be obtained using the average thickness of the crust (17.5 km) and an estimate of the Earth's diameter (12,750 km) (Wikipedia Contributors, 2020; USGS, 2019)
- Crude oil is purely used for automotive gasoline (with a 40% conversion rate) (Freudenrich, 2001)
- Vehicles only use gasoline as their fuel source
- The number of vehicles on Earth right now (~1.645 billion) will remain constant (Hedges, 2025)
- Vehicles drive 10,000 miles per year
- All vehicles have fuel efficiency of 25 miles per gallon

These assumptions are awfully naïve. Assuming that the whole planet's crust is made of oil and that it can be extracted entirely are among several of these claims that are dangerously strong. In addition, most of these figures have been rounded significantly, which will generate even more error in the calculations. Thus, this preliminary estimation will contain a *significant* error term.

The calculations begin with estimating the volume of the Earth's crust. Once this figure is defined and converted to barrels, we determine how many gallons of gasoline each barrel of oil will yield after refinement, resulting in the total number of gallons of gasoline available for use. After that, we estimate how many gallons of gasoline are used per year by multiplying the number of vehicles by the amount of distance driven per vehicle per year and then dividing by fuel efficiency. Finally, dividing the total number of gallons of gasoline by the yearly consumption of gas produces the initial estimation: approximately 1,431,319,316.93 years (~1.43 billion) (see Appendix, Figure 1). This figure is surprisingly large considering that, even without acknowledging the damaging effects of natural gas consumption on the environment, it is arguable that the Earth may no longer be inhabitable after that amount of time (Wikipedia Contributors, 2019).

The predicted longevity of the supply of oil is largely due to the erroneous assumption about the Earth's crust. This is a gross overestimate of how much oil exists and is available for refinement. Although some of our other assumptions are fighting against overestimation, like every vehicle running solely on gas for the rest of time, the claim about the available supply of oil is likely far more impactful on the estimation than the forces in the other direction. Thus, we expect that, in refining our model, the predicted lifespan of crude oil will decrease dramatically in comparison to this initial estimation.

Part Two: Refining the Model

Given the preliminary analysis of the problem, we can now address the assumptions made in part one. The three major aspects that first demand improvement are the estimation of available crude oil and how much gasoline it yields, the lack of consideration for variable fuel efficiency, especially over time, and how many cars and other vehicles there are, also in relation to time. The first part of this section will refine the prediction of total gasoline available for consumption. Considering how the Earth's crust is very far from being purely made up of crude oil, we expect that refining the estimation will lead to a drastic decrease in the lifespan of crude oil, perhaps by an entire order of magnitude. The second part of this section will allow us to account for variable fuel efficiency. Given the increasing popularity of electric vehicles and constantly improving fuel efficiency of gas-powered vehicles, we expect refining this aspect will have a lengthening effect on the lifespan of crude oil. Finally, the third section will address the assumption that there is a constant number of only gas-powered vehicles. This section will model how different types of vehicles become more or less popular over time and how trends in public transportation may vary with time, especially as the world population continues to grow. We expect the improvements made in this section to lengthen the estimated lifespan of crude oil too, as the assumption made in part one did not account for the increasing popularity of electric vehicles, among other things.

Crude Oil Reserves and Refinement

The assumption made in part one regarding the supply of oil is a gross overestimate but also fails to acknowledge other nuances pertaining to recovery and refinement. The best estimates available today predict that there was approximately 6 trillion barrels of total oil existent in the Earth's crust (Oil and Gas Reserves, 2025). Of that, only about half of this is recoverable, which is split in half again due to the refinement process for gasoline. This number is further reduced considering the substantial amount of oil that has already been consumed, about 1.3 trillion barrels (Geiger, 2019). The effect of these reductions is displayed below in figure 1.

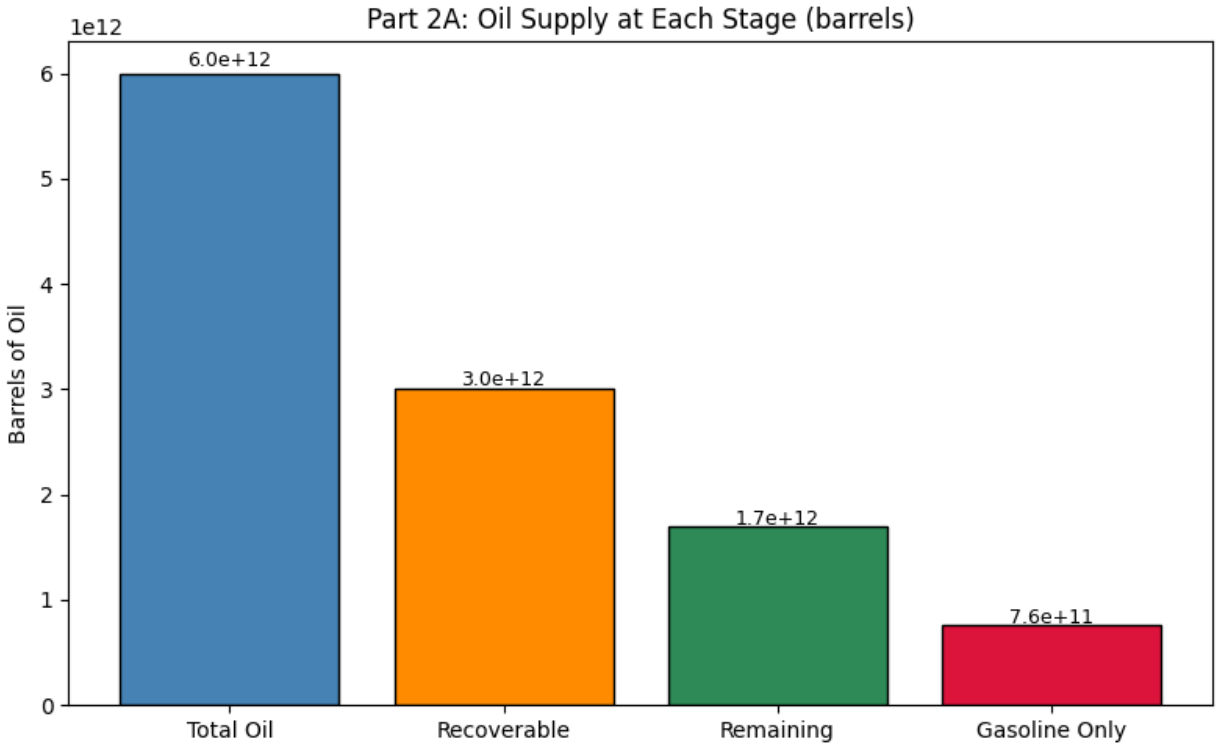


Figure 1: Total Barrels of Oil and Gas at Each Stage

Converted to gallons, our new estimate of total gasoline remaining today is about 32 trillion gallons (see Appendix, Figure 2). This is an astonishing difference from the original prediction, which estimated total gallons of gasoline available to be about 9.42×10^{20} gallons.[1] This decrease of seven orders of magnitude is far greater than we had hypothesized and makes a massive difference on the lifespan of oil. Given the rate of consumption predicted in part one, implementing this new estimate of oil reserves leaves us with only ~55.55 remaining years of oil use. Assuming nonnegligible oil consumption began sometime in the late 19th century, this would imply that the total lifespan of crude oil is just two centuries. In comparison to the original prediction (~1.43 billion years), this improvement has made an enormous impact on our estimation.

Though the total supply of oil reserves has been significantly improved, it still makes some assumptions and thus, still contains some degree of error. Most importantly, the estimations of total crude oil are based on what we know about the Earth, but this information is not exhaustive and may be faulty. The recoverability factor, determined by physical, economic, and technological constraints, may also be incorrect. Improvements in these areas in the future may yield a greater amount of recoverable oil, which would have a noticeable impact on the overall estimate. Furthermore, since the amount of oil previously consumed is also an estimation,

[1] This number represents the total gallons of gasoline predicted in part one, but since it was such a gross overestimate, subtracting out the 1.3 trillion barrels of crude oil that we have already used still leaves us with a number of gallons that rounds to 9.42×10^{20}. This alone displays how incredibly erroneous the previous assumption regarding accessible oil reserves was.

the error contained in that term will inherently show up in our predictions as a result. Despite the effects of each of these assumptions, our improved estimate of oil reserves will serve as a significantly more accurate basis for the rest of the model.

Variability in Fuel Efficiency

In part one, we assumed that every vehicle drove 10,000 miles per year and that they all had the same fuel efficiency of 25 mpg. This is far from reality and fails to recognize the enormous variation that truly exists. To improve upon this assumption, we split the total number of vehicles into seven distinct categories being SUVs, semi-trucks, transit buses, motorcycles, cars, pickup trucks, and electric/hybrid vehicles. These categories each contain much less variation in fuel efficiency and yearly mileage than compared to before, which will improve the overall accuracy of predicted gas consumption over time. In addition, this section models the change in fuel efficiency over time for each of the categories, as new technology continues to improve efficiency for most of these vehicles, but at varying rates. The model of changing fuel efficiency by category is shown below in figure 2.

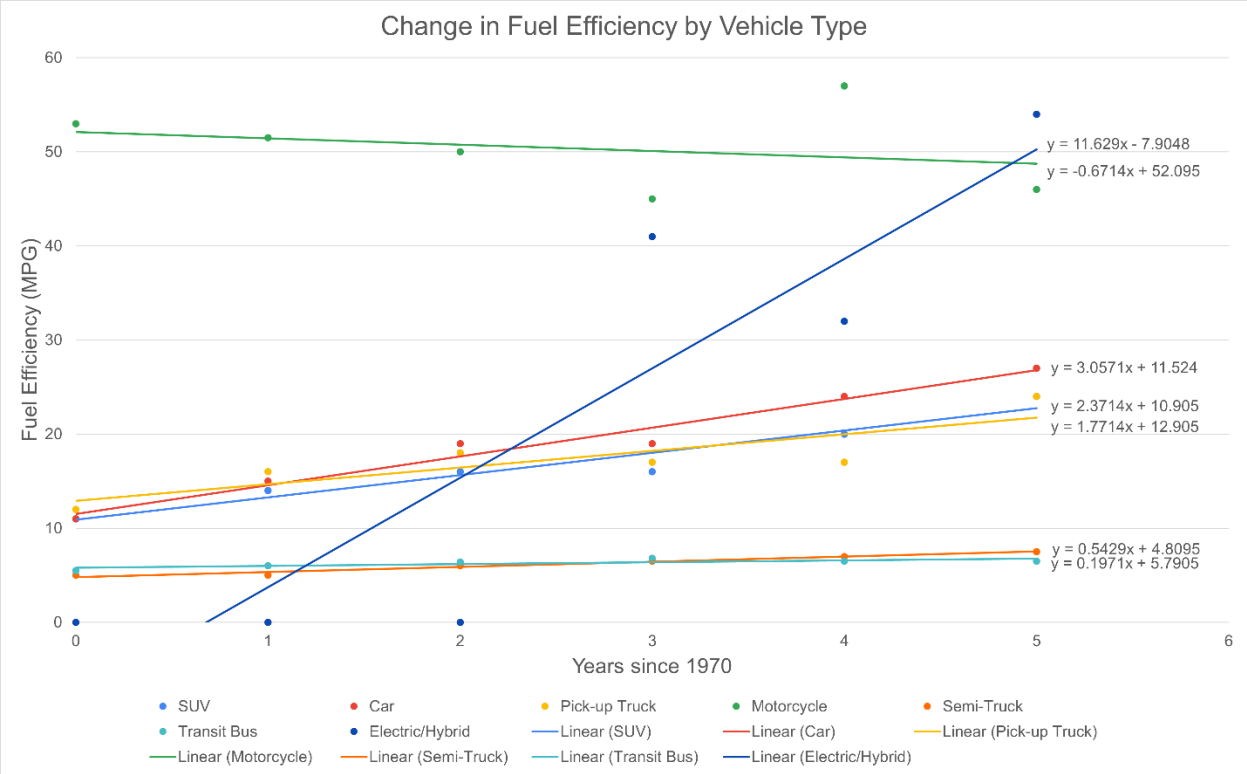


Figure 2: Change in Fuel Efficiency by Vehicle Type Over Time

This chart displays vastly different fuel efficiency between categories, as well as the overall improvements over time. Although the relationships seen on the graph seem to be represented well using linear regression, it is unrealistic that fuel efficiency will continue to improve at a constant rate; that is, linear growth becomes increasingly unrealistic as time goes on since fuel efficiency is bounded by physical limitations. Regardless, applying the dynamic fuel efficiency

model to our total gas estimate from part one, we get a new estimated lifespan of about 5.7 billion years. This agrees with the hypothesis that modeling fuel efficiency would increase the total lifespan of crude oil.

As mentioned, fuel efficiency cannot grow linearly. This assumption causes fuel efficiency to get very high in the future, meaning that gas consumption will approach zero, which is unrealistic. Although, this model also assumes that there is a constant proportion of electric vehicles. Since electric vehicles use much less crude oil and are rapidly increasing in popularity, not accounting for this will, to some extent, counteract the effect of assuming linear growth in fuel efficiency. More importantly, these assumptions are both increasingly problematic further in the future, but since the advancements in section 2A have drastically decreased how much gas is available in the first place, this will inherently decrease how far into the future we will have to predict. In addition, the next section will address changes in the number of electric vehicles over time, so once all three sections are combined in part three, the assumptions made in this section will have much less of an impact on the accuracy of our overall estimation.

Variability in Total Vehicles and Nuances in Types

In order to estimate the number of vehicles over such a long time period, finding a correlated variable that is easier to predict helps improve accuracy. We see that, given the US vehicle and population data, these two variables have a correlation coefficient of about .935, a very strong relationship. Despite this high correlation, using population as an explanatory variable creates an issue since there is quite a bit of uncertainty regarding the carrying capacity of humans on Earth. Thus, we will assume it is around 11 billion (somewhat of a middle ground regarding our best estimates) but slowly decreasing over time. The long term decrease not only helps to address diminishing resources but also helps to decrease the estimated number of vehicles, as such a large amount of people likely exists alongside more urban life and thus more shared transportation. The long-term prediction of population is visualized below in figure 3, with historical data and official estimates up to 2100 laid on top of our logistic model (United States Census Bureau, 2026).

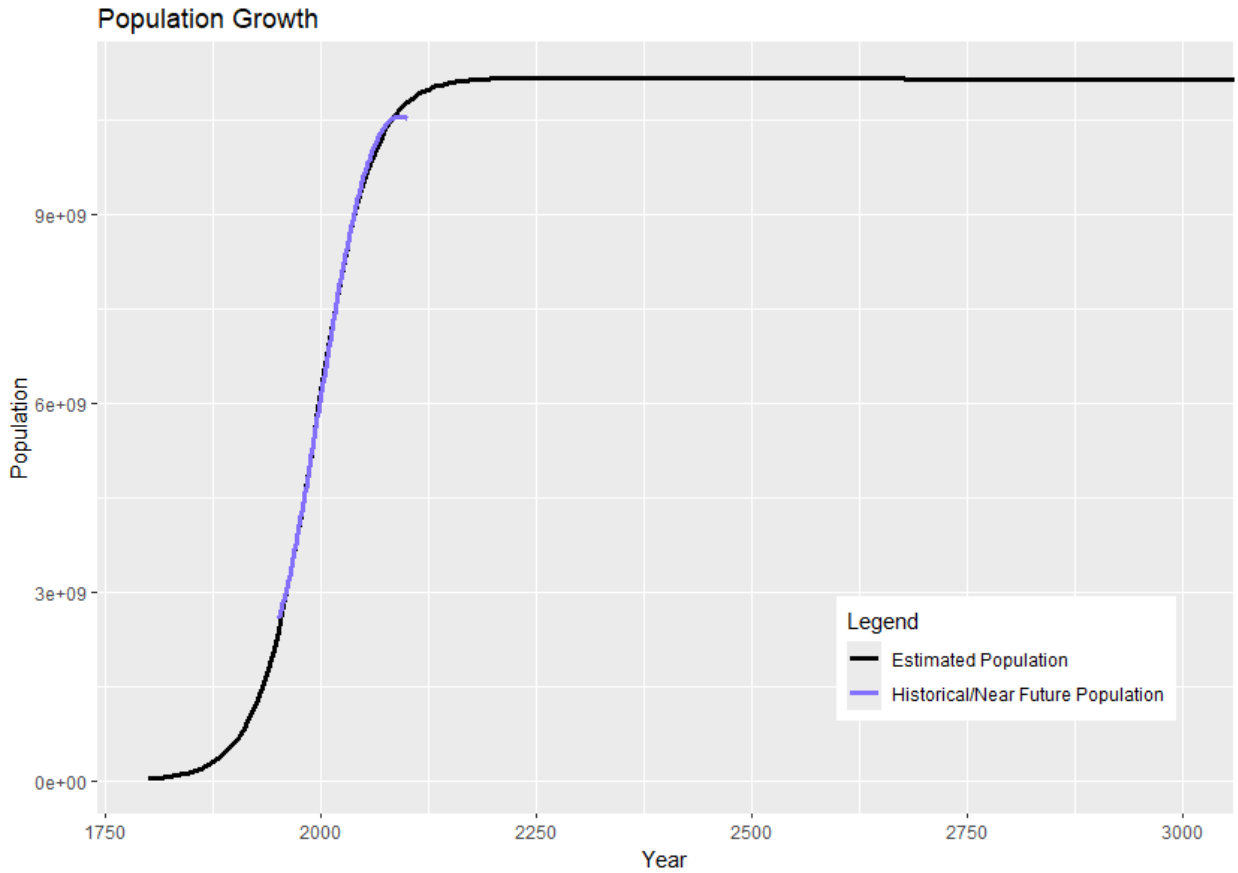


Figure 3: Historical and Estimated World Population Growth

The number of vehicles in the world is far from constant and additionally, electric vehicles have been rapidly gaining popularity, which is completely ignored in part one. States like California have already passed legislation that expects to phase out combustion vehicles completely within a few decades (California Air Resources Board, 2022). Although California is ahead of the curve in this regard, the shift towards electric vehicles is a global trend. Thus, this model will incorporate the growing popularity of electric vehicles by maintaining the rate of change of the proportion of electric vehicles to total vehicles registered in the US based on the past few years (US Department of Energy, n.d.). This is an assumption that is fairly reasonable to make, although could be potentially misrepresentative of the future. Electric vehicles may be adopted sooner or later than expected, or at differing rates over time. Tractor trailers may need much longer to convert to electric power. It is hard to know the answers, but this assumption should still lead to a fairly accurate prediction. Figure 4 displays this exchange, demonstrating a complete swap just before 2300. Electric vehicles still use crude oil to produce the energy they run on, but since they are significantly more efficient, this will drastically decrease how much crude oil is needed to power vehicles in general (Kirk, 2024).

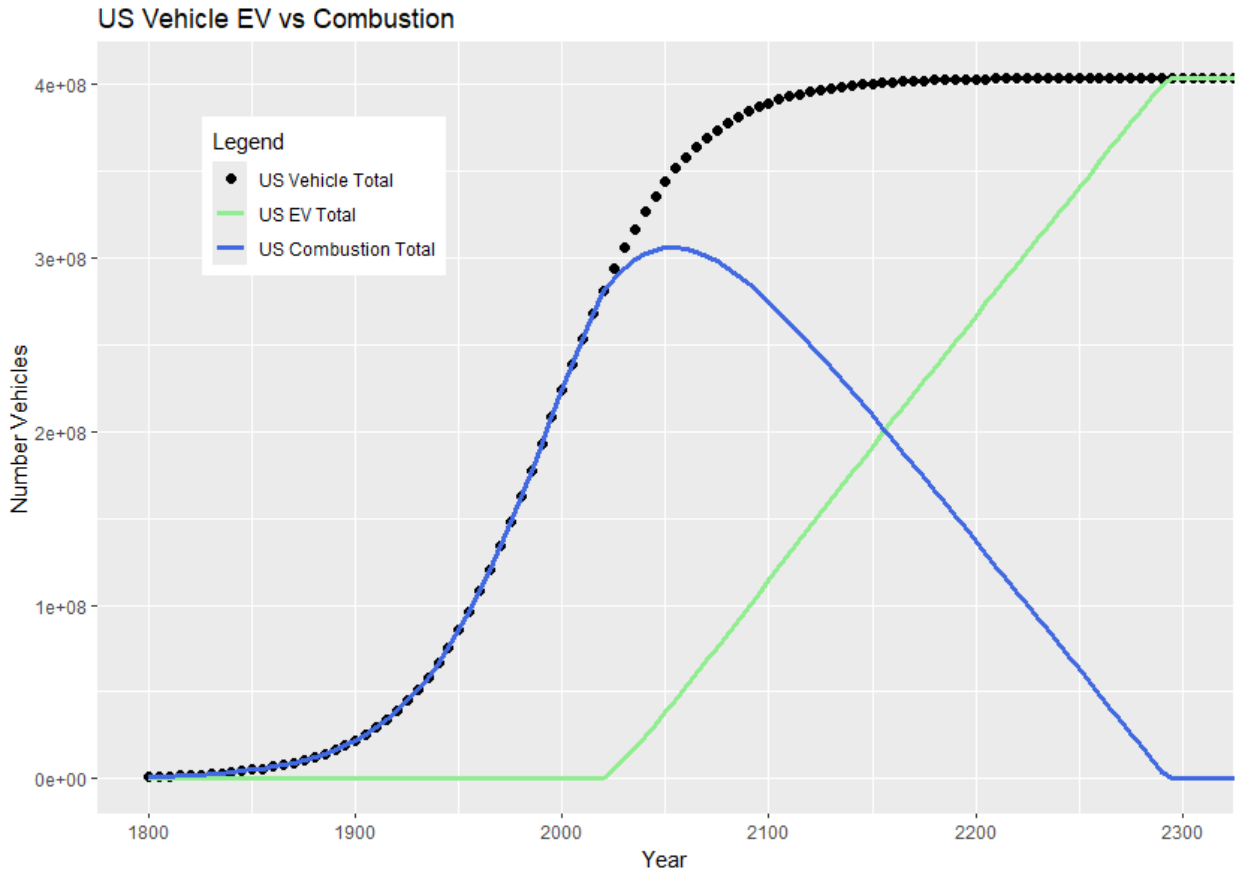


Figure 4: Estimated Growth of Total Vehicles and the Swap from Combustion to Electric

Aside from the swap from combustion to electric, the improvements made in this section also divide the total number of vehicles on the road up into six major categories, in the same way the Bureau of Transportation Statistics (BTS) splits up highway registered vehicles. The categories are light-duty vehicle (short wheelbase), motorcycle, light-duty vehicle (long wheelbase), truck (single-unit 2-axle 6-tire or more), truck (combination), and bus (Schmitt et al., 2025). By separating out these categories, we can better model fuel efficiency since there is quite a significant difference between motorcycles and tractor trailers, for example. Again, some assumptions must be made to proceed. Due to a lack of reliable international vehicle data and a relatively small number of years of data within the US, we will assume that the rest of the world has the same proportion of these categories and that these proportions will remain constant. We will also assume that the proportion of vehicles in the US compared to the rest of the world will stay constant, using estimates made for 2024 (Hedges, 2025). Due to these assumptions, although figures 4 and 5 show the change in vehicles in the US over time, it is inherently the same relationship that occurs on the global level, just on a different scale.

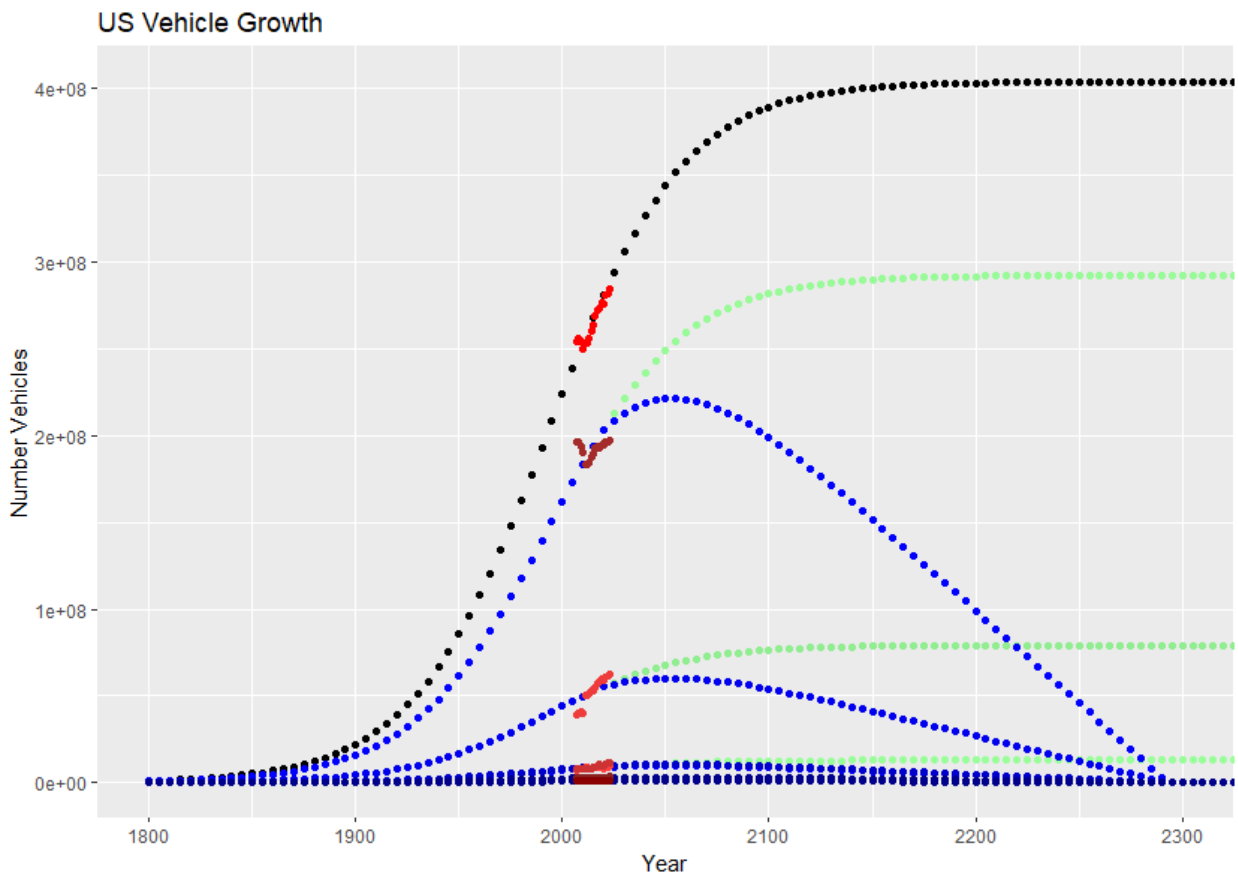


Figure 5: Estimated Growth in Total Vehicles in the US, Broken into Six Categories

In figure 5, we see the estimated change in each category of vehicles (blue/green) and in aggregate (black) compared to the actual data collected by the BTS (red). The blue represents total number of each category of combustion powered vehicles and the switch to green represents the total number of each of these vehicles once electric vehicles were introduced. Over time, we see that the total number of vehicles in the world (not visualized on these plots) tops over two billion before slowly retreating (see Appendix, Figure 3). This is close to a 50% increase compared to the constant assumed in part one, a significant difference affecting our overall estimation.

With an improved model of how the number of vehicles changes over time, we can update our estimate of the lifespan of oil. The model is altered to make 1,000-year time steps, instead of five-year time steps, to improve efficiency. This yields a new estimate of ~4.23 billion years, almost three times longer than the original estimate made in part one. This result agrees with our hypothesis, which expected that a more accurate model of the number of vehicles over time would increase the lifespan of crude oil, but not to the extent of a whole order of magnitude.

Part Three: Combining the Models

The three major assumptions that were addressed in part two each had a significant impact on the total estimated longevity of crude oil. Since these assumptions were addressed in isolation, the resulting changes indicate that each of the respective assumptions did indeed introduce a fair amount of error to our basic model in part one. Thus, implementing all three of the improvements together should yield a much more accurate estimation than part one. We expect that the cumulative effect of the three changes will be a shortened lifespan of crude oil by several orders of magnitude, mainly due to the huge decrease in available oil seen in section 2A, along with insight into how consumption might change over time.

To start integrating the new improvements into our model, the new estimation for total gasoline had to be substituted for the old one. The only issue was that the new estimation represents how much gasoline remained in 2020, whereas the original estimation represented the total supply of gasoline. Thus, the data for the other two sections had to be truncated to ensure the proper alignment of time when calculating our final estimation. With this set, the next task was to combine the fuel efficiency model with the vehicle count model. Since these two models used slightly different categories, the ones from section 2B were mapped to the most reasonable fit in section 2C. All the categories in section 2C were assigned exactly one category from section 2B except for *light vehicle, long wheelbase* and *truck (6+ tires)*. SUVs and pickup trucks both have fuel efficiency ratings that put them closest to *light vehicle, long wheelbase*, so their fuel efficiency was applied to this new category in a proportional manner (based on the vehicle totals found in section 2B). Since none of the section 2B categories reasonably applied to *truck (6+ tires)*, this category was best represented by constants. This is less of an issue than before, though, because we now have separate categories, which allows us to choose constants that better represent this specific category's yearly mileage and fuel efficiency. In addition, this category represents a relatively small portion of the total number of vehicles, making its impact on overall accuracy even smaller. With the categories properly mapped to fit the section 2C model, we then created a few new variables in the data frame to represent yearly gas consumption per category and then summed them up to get total yearly gas consumption, which is displayed below in figure 6.

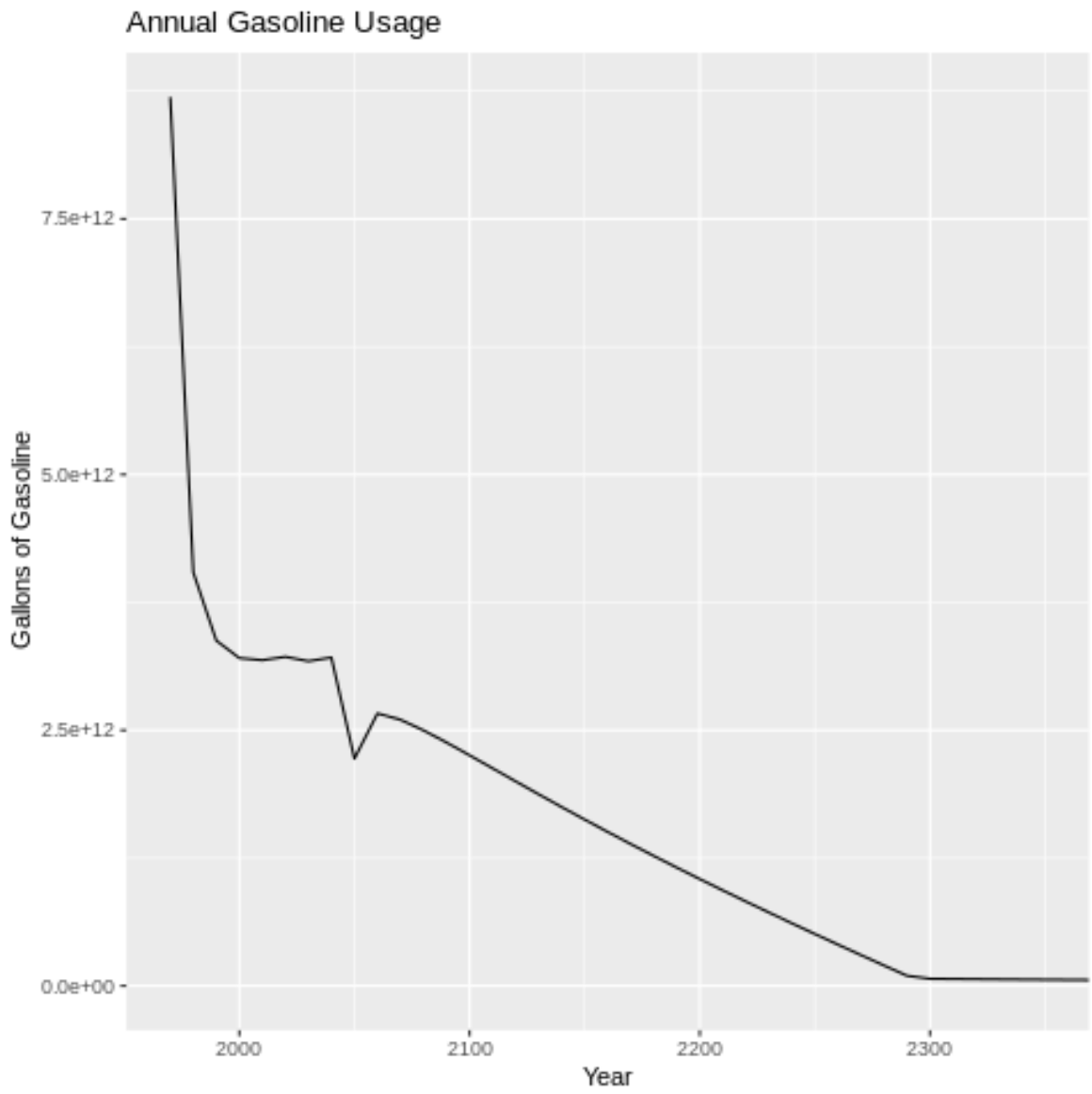


Figure 6: Total Annual Gasoline Usage (1970-2350)

The odd behavior seen around 2050 is due to the switch from combustion powered vehicles to electric vehicles interacting with the changes in fuel efficiency.

With a few more minor changes to the code, the three sections from part two were all aligned based on time, allowing us to determine how many years of gasoline use remained given our improved estimations. The new and improved estimation is, starting in 2020, about 130 remaining years (visualized below in figure 7).

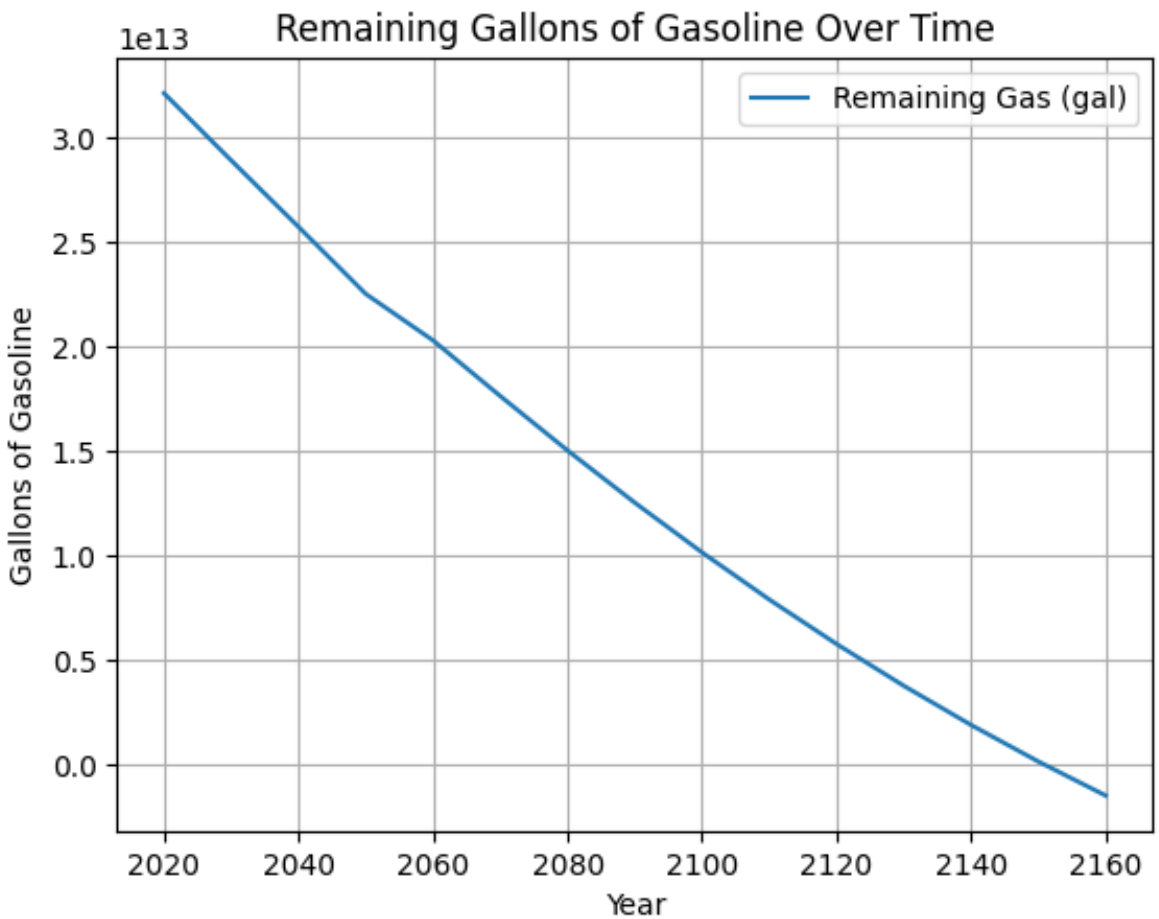


Figure 7: Estimated Remaining Gasoline Reserves

Assuming nonnegligible oil consumption began sometime in the late 19th century, this suggests that the total lifespan of crude oil based on gasoline consumption is just shy of three centuries. The drastic difference between this and the estimation made in part one is far greater than we originally anticipated. It is a difference of seven orders of magnitude and, more importantly, suggests that running out of oil is an imminent concern.

A surprising result of the work done thus far is that our improved model suggests oil will run out at almost the exact same time as it expects electric vehicles to outnumber combustion vehicles. Though our model still relies on plenty of assumptions, it does line up with similar work done by professionals in the field. In fact, our model is arguably generous in its estimation. With that said, this observation regarding electric vehicles suggests some possible preliminary elements for a solution to running out of oil. Electric vehicles make far more efficient use of crude oil, which, in addition to helping mitigate certain environmental concerns, could allow us to prolong the lifespan of crude oil by a significant amount. The plot shown below in figure 8 is a rough visualization of what it might look like if we were 50 years ahead in terms of progress

towards being fully electric. It displays a hypothetical remaining lifespan of oil (700 years) that is five times longer than before.

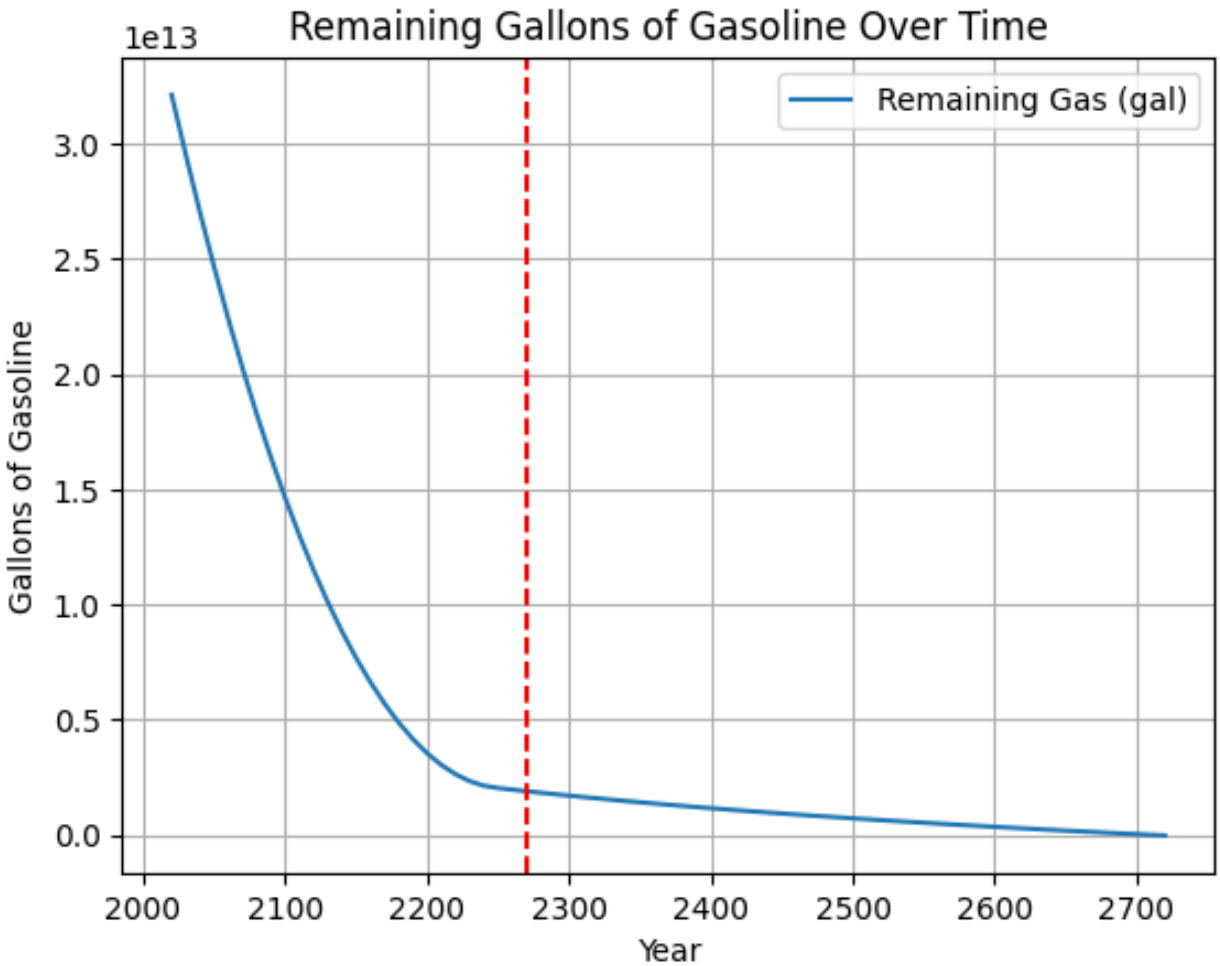


Figure 8: Adjusted Estimated Gasoline Reserves

Although this does inherently benefit from 50 years of fuel efficiency improvements as well, the contribution of that is almost negligible compared to the effect of going fully electric. The red dashed line displays the point at which our model expects more than 90% of all vehicles to be fully electric. This is right about when the curve dramatically flattens out. It is no coincidence that the area under the curve after this point is far less than before, as this area represents aggregate oil consumption and displays how much more efficient electric vehicles truly are.

Part Four: Further Refinement

Aside from what we've already considered, there are many more factors that influence the lifespan of crude oil. This section intends to explore a few of these other factors and then analyze the sensitivity of our model by adjusting certain parameters to see how they will affect the results.

Economic Factors

Economics can provide a unique insight into the dynamic consumption of crude oil over time. This section aims to uncover how price relates to demand, and thus consumption. We expect the changes made in this section to decrease the overall lifespan of crude oil, since inflation in gas prices is surprisingly low compared to other goods, which will likely cause demand, and thus consumption, to remain elevated in the short term.

After cleaning and combining consumption and price data, we can analyze possible relationships through a correlation table, which reveals that time is a strong predictor of the log of nominal price, which is, in turn, correlated with the ten-year change in consumption (see Appendix, figure 4). These relationships are visualized in figure 9.

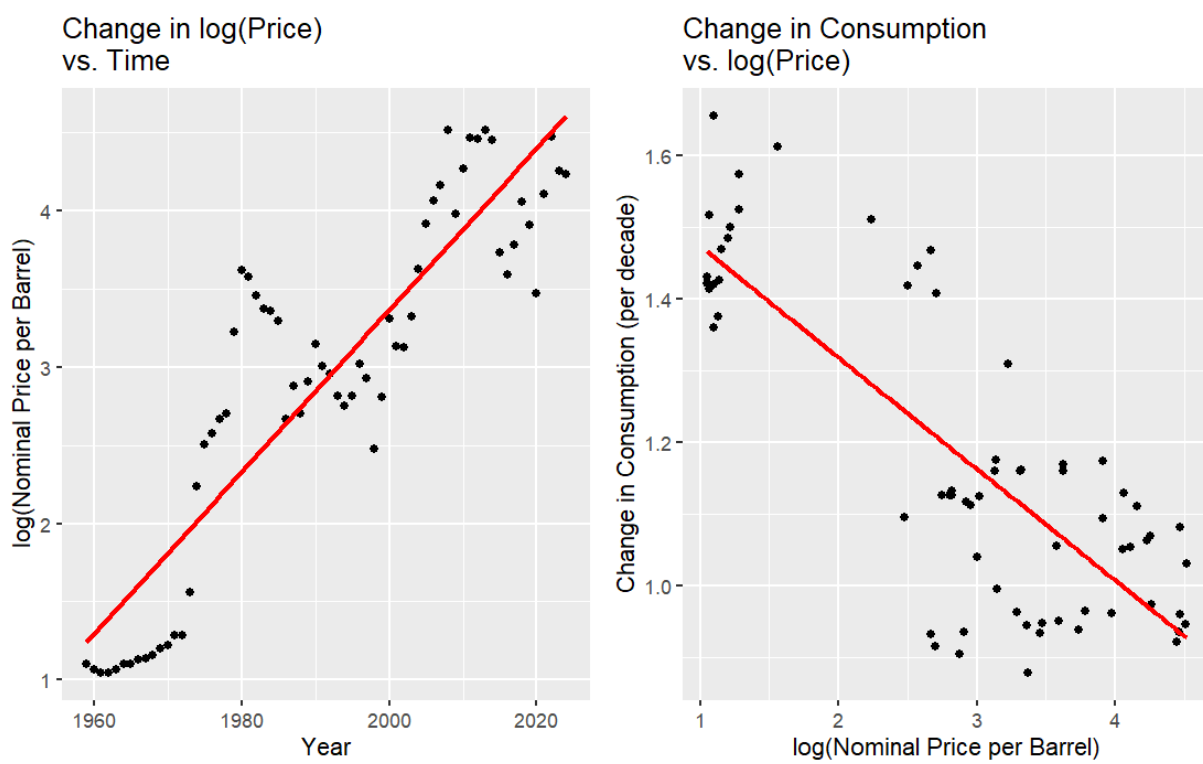


Figure 9: Economic Regression

Since this is a relatively good fit, using it to predict the impact of economic price on demand and thus consumption is reasonable in the short term.[2] After doing so, we adjust our prior estimation for annual gasoline usage, which is shown below in figure 10.

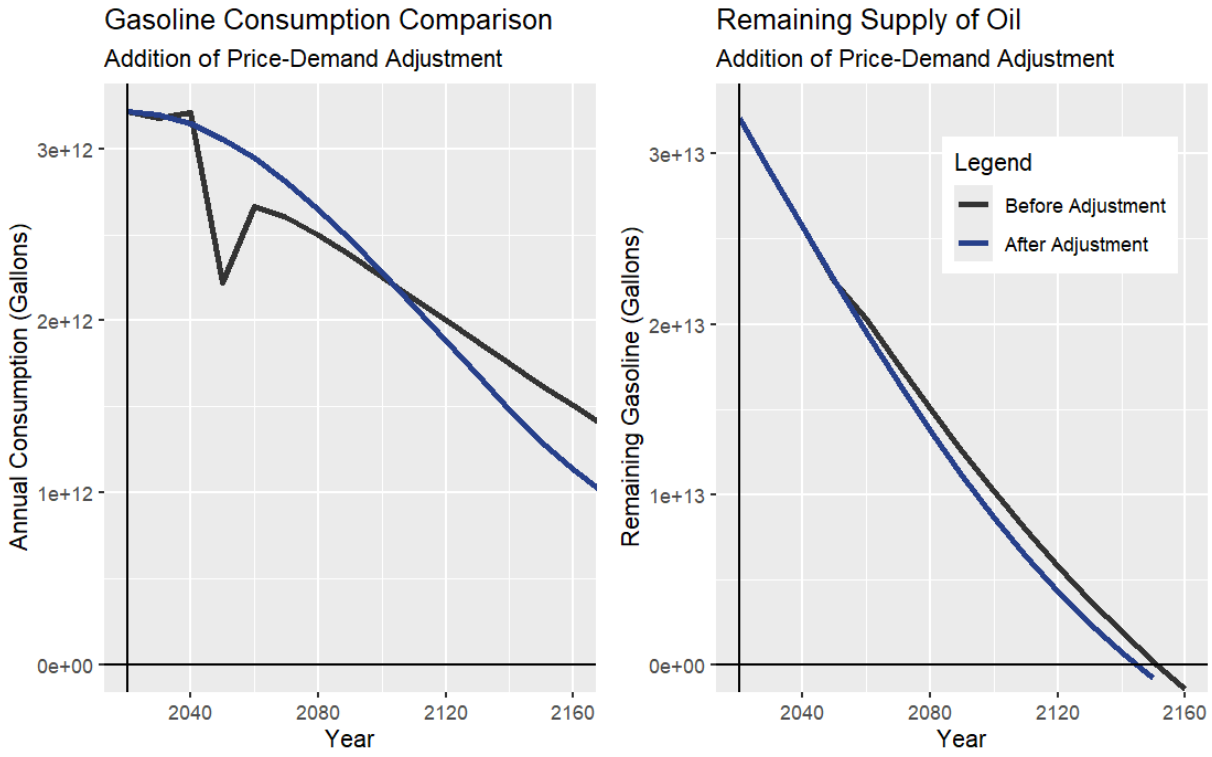


Figure 10: Impact of Price on Consumption

As expected, consumption is slightly inflated in the near term, but lower in the long term. The short-term impact is fairly large, though, leading the model to predict a slightly shorter lifespan than before of about 125 remaining years.

This adjustment has only looked at the impact of time on price and price on demand, although there are many other economic factors that could be used to predict consumption, like total oil rigs online and industrial activity. Surprisingly, these other factors weren't statistically significant when included in the regression models, which is why more variables aren't included.

Political Factors

Political factors are closely tied to economic factors but play a distinct role in crude oil consumption. Considering how the world economy is built on crude oil, it makes sense that this is a deeply political issue. This is an incredibly complex aspect of crude oil consumption as political factors include everything from environmental policy to geopolitical tension to so much more. Accounting for all these different factors is very difficult, and predicting them is effectively impossible, but there are some key trends that suggest how they may impact oil

[2] In the long term it overestimates the impact on demand, so a discounting factor was included to account for this in later years.

consumption in the future. Namely, there is a clear pattern of market-driven decision making that heavily influences policy decisions. Growing international competition often pushes countries to set aside environmental initiatives to stay competitive in the global economy, which suggests that demand may remain elevated for a long time, drawing up total consumption. There is often a lot of pushback in this regard, though, as environmental policy is just as significant as a concern, so it is hard to tell which direction politics will go and how this will affect crude oil consumption in the future.

Other Types of Vehicles

In part two, we split up the generalization about annual miles driven and fuel efficiency into several different categories, which helped provide a more detailed and nuanced understanding of crude oil consumption over time. There are still many more categories of vehicles that haven't even been considered, though, many of which consume enormous amounts of crude oil. Some examples include airplanes, boats, and vehicles that consume fuel other than gasoline (i.e. diesel), as well as certain construction and agricultural equipment. To account for these other vehicles, we could add extra categories to our model that would bulk up predicted annual consumption. This would almost certainly decrease the total lifespan, and by a fairly significant amount. This is probably the largest factor still missing from our model and would explain why our prediction is still almost double the length of certain experts.

Other Predictive Modeling Techniques

In addition to adding detail to the model, structural changes may also make a meaningful impact on our prediction. That is, several variables were predicted using linear regression, which is exceedingly unrealistic in the long term. This wasn't much of an issue after the changes made in section 2A since the lifespan was brought down so low, but the problem with linearity likely is still causing significant overestimates. For example, in section 2B we estimate linear growth in fuel efficiency over time, but the changes made in this regard were much more rapid in the past few decades than they reasonably could be moving forward. We also modeled the growth of electric vehicles using a linear relationship, which is unrealistic as well, especially for vehicles like long-haul semi-trucks. Switching to a logistic model or machine learning based algorithms for these variables would be a lot more realistic in later years, as they would better capture nonlinear growth. This would likely increase consumption, as it is letting inefficient vehicles affect the model further into the future. Thus, improving the modeling techniques would most likely decrease the overall prediction for the remaining lifespan of crude oil.

Sensitivity Analysis

Although this project has addressed many different factors that determine the lifespan of crude oil, the estimations made are all based on the same constants that were established in part 2A. Thus, analyzing the model's sensitivity to changes in parameters is important. Sensitivity

analysis will reveal how stable this model is, but it will also provide a better understanding of the lifespan of crude oil by creating a range of reasonable values instead of a single number.

The parameters that lie beneath all the calculations in this model are *total oil*, *recovery factor*, *used oil*, and *gas yield*. Since each of these factors are subject to varied levels of uncertainty, the percent change tested was not the same across the board. For example, the recovery factor is highly variable due to factors like geopolitical tension, profitability, regulation, and more. This variable was adjusted up and down by 15%, which should account for a reasonable level of variation that we may have failed to capture regarding these uncertainties. In contrast, the post-refinement gasoline yield is a number that is well studied, so we can be relatively certain that it is accurate. Furthermore, it seems unlikely that there will be any major advances of technology over the course of the predicted lifespan that will drastically increase gasoline yield, so this parameter was only adjusted by ±5% in the sensitivity analysis. The results of the parameter sweep are shown in figure 11.

Parameter	Adjustment (Percent)	Estimated Years Remaining	Change from Baseline (Years)	Change from Baseline (Percent)	Elasticity[3]
Baseline	-	124.82	-	-	-
Oil in Place	±10%	(93.29, 178.01)	(-31.53, 53.20)	(-25.26%, 42.62%)	(2.53, 4.26)
Recovery Factor	±15%	(80.57, 232.70)	(-44.24, 107.88)	(-35.45%, 86.44%)	(2.36, 5.76)
Used Oil	±5%	(116.93, 133.55)	(-7.88, 8.73)	(-7.88%, 8.73%)	(-1.40, -1.26)
Gas Yield	±5%	(114.68, 136.46)	(-10.13, 11.65)	(-8.12%, 9.33%)	(1.62, 1.87)

Figure 11: Sensitivity Analysis

We see that the most sensitive parameters were oil in place and the recovery factor, which makes sense as they are the basis of the lifespan calculation. They both have elasticity of about 2.45 in the negative direction, but a much greater elasticity of 4.26 (oil in place) and 5.76 (recovery factor) in the positive direction. This is because the model is using a decreasing amount of crude oil as time goes on due to the increasing popularity of electric vehicles. Thus, adjusting these parameters up results in an even longer extension of the crude oil lifespan relative to the original parameter since the oil is being used more efficiently further in the future and being demanded less due to increasing prices.

Aside from the adjustments discussed in this section, there are some other factors that would be interesting to test too. This model relies heavily on U.S. data since it has been fairly well kept at a detailed level for quite some time. Thus, we can carry out predictive modeling with this data, but extrapolating this to the rest of the world introduces a fair bit of potential error. It is unlikely that the ratio of vehicle types in the U.S. is perfectly representative of the rest of the world, and it is also unlikely that the ratio of vehicles in the U.S. to the rest of the world will

[3] $Elasticity = {}^{\%\Delta\ lifespan}/_{\%\Delta\ parameter}$

remain the same in the future. This was an assumption we had to make, but testing to see how sensitive it is would reveal how strong the assumption truly is.

Despite the countless different parameters we could have tested, the sensitivity analysis is nonetheless insightful. It shows that our prediction isn't unstable, but it does reveal a fairly wide window of possible lifespans. This is to be expected, though, as there are countless different variables that affect the lifespan of crude oil, and accounting for them all is unrealistic.

Conclusion

Estimating the lifespan of crude oil is quite a substantial undertaking. Considering the vast number of factors that are constantly changing, just getting a reasonable estimate takes a lot of care. This was abundantly evident in section 2A, as we brought our prediction down from a number in the billions to just over 55 years, a change largely due to dealing with an excessively strong assumption. The further refinements made in part two added quite a bit of detail and meaning behind the overall estimate too. Once combined in part three, our estimate landed around 130 remaining years, as introducing varying fuel efficiency and increased popularity in electric vehicles lengthened our prediction.

After several phases of refinement and analysis, our final prediction window was approximately 80 to 230 remaining years, with the particular prediction lying around 125 remaining years. This isn't too far from some professional estimations and shows that, despite its sensitive nature, our model is relatively robust, especially compared to where it started in part one. In addition, the final prediction was not very different from where we landed in part three, implying that the most important refinements were made in part two and that further work is mostly just reducing sensitivity rather than increasing accuracy.

Group Evaluation

As a team, we did a good job combining our varied skillsets in a way that was effective. There were many technical challenges due to different technologies used and remote collaboration, but we were able to communicate well and work through these problems with ease. The work for this project was split up roughly as follows:

Evan Scheuermann:

- Project write-up (each member was responsible for their own write-up)
- Part one—development and implementation (python)
- Part two—general planning
    - Section C (Variability in Total Vehicles and Nuances in Types)—research, development, and implementation (R)
- Part three—planning and integration (combined python/R implementation)
- Part four—planning, economic factor research, development, and implementation, sensitivity analysis

Jake Scheidelman:

- Part two
    - Section A (Crude Oil Reserves Refinement)—research, development, and implementation (python)
- Part three—planning and collaboration
- Other supporting contributions

Lucy Haynes:

- Part two
    - Section B (Variability in Fuel Efficiency)—research, development, and implementation (excel and python)
- Part three—planning and collaboration
- Other supporting contributions

References

California Air Resources Board. (2022, August 25). *California moves to accelerate to 100% new zero-emission vehicle sales by 2035 | California Air Resources Board*. California Air Resources Board. https://ww2.arb.ca.gov/news/california-moves-accelerate-100-new-zero-emission-vehicle-sales-2035

Freudenrich, C. (2001, January 4). *How Oil Refining Works*. HowStuffWorks. https://science.howstuffworks.com/environmental/energy/oil-refining1.htm

Geiger, J. (2019, August 20). *How Much Crude Oil Has The World Really Consumed?* OilPrice.com. https://oilprice.com/Energy/Crude-Oil/How-Much-Crude-Oil-Has-The-World-Really-Consumed.html

Hedges, J. (2018). *US VIO Vehicle Registration Data 2018, Fast Quote on Car Data*. Hedges & Company. https://hedgescompany.com/automotive-market-research-statistics/auto-mailing-lists-and-marketing/

Hedges, J. (2025, October 26). *How many cars are there in the world in 2021? Stats by country*. Hedges & Company. https://hedgescompany.com/blog/2021/06/how-many-cars-are-there-in-the-world/

Kirk, K. (2024, January 29). *Electric vehicles use half the energy of gas-powered vehicles*. Yale Climate Connections. https://yaleclimateconnections.org/2024/01/electric-vehicles-use-half-the-energy-of-gas-powered-vehicles/

Oil and Gas Reserves. (2025). Penn State College of Earth and Mineral Sciences. https://courses.ems.psu.edu/earth104/node/1298

Online Conversion - Volume Conversion. (n.d.). Www.onlineconversion.com. https://www.onlineconversion.com/volume.htm

Schmitt, R., Shaw, J., DiJoseph, P., & Stacy, M. (2025). Transportation Statistics Annual Report 2025. *ROSA P*. https://doi.org/10.21949/w2zr-3a26

U.S. Energy Information Administration. (2022, June 1). *U.S. Crude Oil First Purchase Price (Dollars per Barrel)*. Www.eia.gov. https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=pet&s=f000000__3&f=a

U.S. Energy Information Administration. (2026, April). *Monthly Energy Review, Table 3.1*. U.S. Department of Energy. https://afdc.energy.gov/data

United States Census Bureau. (2026). *International Database*. Census.gov. https://www.census.gov/data-tools/demo/idb/#/dashboard?dashboard_page=country&COUNTRY_YR_ANIM=2026

US Department of Energy. (n.d.). *Alternative Fuels Data Center: TransAtlas*. Afdc.energy.gov. https://afdc.energy.gov/transatlas#/

USGS. (2019). *Inside the Earth [This Dynamic Earth, USGS]*. Usgs.gov. https://pubs.usgs.gov/gip/dynamic/inside.html

Wikipedia Contributors. (2019, April 25). *Future of Earth*. Wikipedia; Wikimedia Foundation. https://en.wikipedia.org/wiki/Future_of_Earth

Wikipedia Contributors. (2020, January 30). *Earth's crust*. Wikipedia; Wikimedia Foundation. https://en.wikipedia.org/wiki/Earth%27s_crust

World Bank. (2025). *Population, Total*. The World Bank; World Bank Group. https://data.worldbank.org/indicator/SP.POP.TOTL

Appendix

1. Calculating an initial estimate under naïve assumptions:

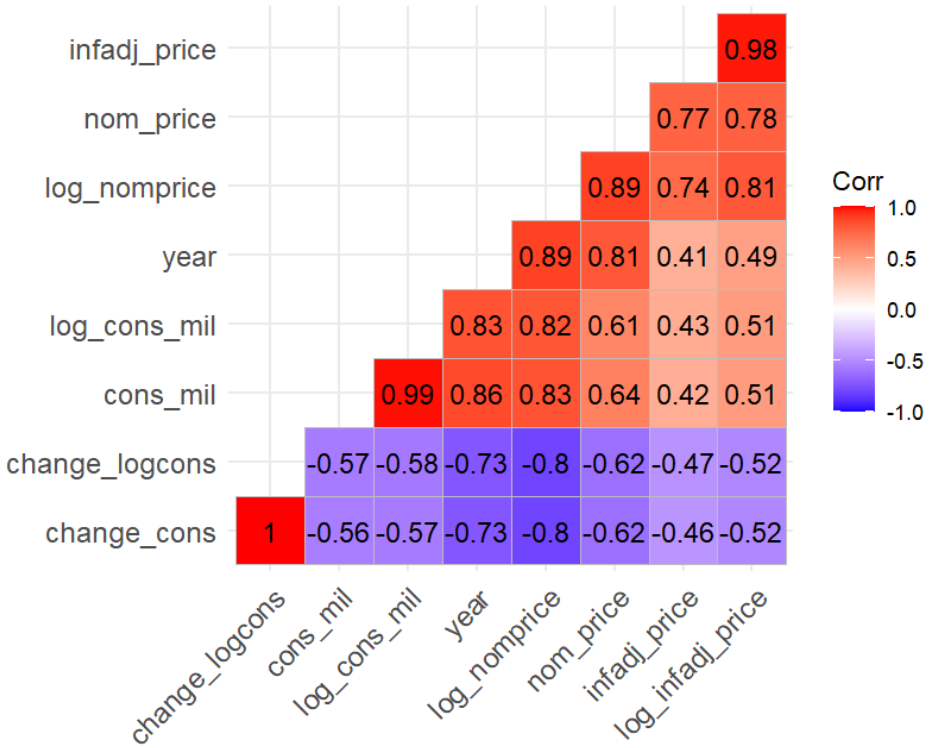
$$Volume_{crust} = \frac{4}{3}\pi\left(\left(\frac{12750}{2}\right)^3 - \left(\frac{12750}{2} - 17.5\right)^3\right) \approx 8.91 \times 10^9\ km^3$$

$$Total\ Gasoline = (Volume_{crust})\left(\frac{6{,}289{,}810{,}770.4\ barrel_{oil}}{km^3}\right)\left(\frac{2\ barrel_{gas}}{5\ barrel_{oil}}\right)\left(\frac{42\ gallon}{barrel}\right)$$
$$\approx 9.42 \times 10^{20}\ gallons$$

$$Yearly\ Consumption = (1.645 \times 10^9\ vehicle)\left(\frac{10{,}000\ mile}{vehicle}\right)\left(\frac{gallon}{25\ mile}\right)$$
$$\approx 6.58 \times 10^{11}\ {}^{gallons}/_{year}$$

$$Crude\ Oil\ Lifespan = (gallons_{total})\left(\frac{year}{gallons}\right) \approx 1.43 \times 10^9\ years$$

2. $Remaining\ Gasoline = 42 * (Oil_{recoverable} - Oil_{consumed}) * yield_{gasoline}$
   $Remaining\ Gasoline_{2020} = 42 * (3 \times 10^{12} - 1.3 \times 10^{12}) * .45 \approx 3.2 \times 10^{13}\ gallons$
3. Sample data displaying the observation year, estimated total number of vehicles in the world, and estimated number of light-duty vehicles (short wheelbase), motorcycles, light-duty vehicles (long wheelbase), trucks (single-unit 2-axle 6-tire or more), trucks (combination), and buses in the world, respectively.

year	world_veh_tot1	sumworld_lts	sumworld_moto	sumworld_ltl	sumworld_6tr	sumworld_tcombo	sumworld_bus
1990	1062349467	768787098	33922723	208466232	36497103	11096057	3580254
1995	1147561649	830452330	36643700	225187530	39424575	11986084	3867430
2000	1232361457	891819140	39351510	241827908	42337879	12871803	4153216
2005	1315772209	952180737	42014965	258195709	45203462	13743014	4434321
2010	1396880583	1010876179	44604901	274111712	47989947	14590177	4707667
2015	1474874842	1067318039	47095397	289416628	50669446	15404814	4970518
2020	1549073826	1121013385	49464704	303976792	53218559	16179809	5220578
2025	1618944887	1171576756	51695812	317687682	55618985	16909600	5456052
2030	1684110470	1218734928	53776666	330475210	57857754	17590244	5675669

4. Economic factors correlation matrix: